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Holocene vegetation dynamics and climate variability inferred from pollen records from Northeast India

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ABSTRACT

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We report the preliminary findings of a palynological investigation conducted on a 50 cm long sediment core retrieved from a swamp within the Gauhati University Campus, Kamrup Metropolitan District, Guwahati, Assam, Northeast India for understanding the vegetation dynamics. The palynological assemblage reveals a rich diversity of pollen taxa including tropical moist deciduous taxa, marshy taxa, highland species, anthropic indicators and ferns. Based on the relative abundances these pollen taxa, three distinct pollen zones GU-I, GU-II and GU-III have been identified. The presence of tropical moist deciduous taxa including *Syzygium cumini, Shorea robusta, Lagerstroemia* in the first zone suggest that a moist and more humid climate prevailed throughout the period of sediment accumulation. The presence of highland taxa such as *Abies*, *Pinus* and *Alnus* indicates wind activity facilitating the transport of pollen grain from higher elevations. Furthermore, the higher frequencies of cultural pollen taxa such as *Poaceae* and *Cerealia* in Pollen Zone GU-I and GU-II reflect significant human activity. The decline of moist deciduous taxa along with non-arboreal elements in Pollen Zone GU-II suggests a shift towards relatively cooler and drier climatic conditions. In contrast, a subsequent increase in the abundance of *Shorea robusta* in Pollen Zone GU-III (younger zone) indicates an amelioration of climate, characterized by relatively warm and humid conditions than previous phase in the region.

Keywords: Pollen, vegetation, climate, Holocene, Assam, Northeast India.

INTRODUCTION

Northeastern India experiences climate primarily shaped by the Indian Summer Monsoon (ISM), marked by intense seasonal rainfall and recurrent flooding (Breitenbach et al. 2010). Studies conducted across the Northeast Indian region have reported substantial alterations in forest composition and structure in response to climatic variability, anthropogenic pressures, and variations in monsoon intensity (Bera & Basumatary 2013). Lakes, functioning as natural sedimentary basins, represent vital repositories of palaeoenvironmental

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records. It is widely recognized that many of these lakes originated during the early Holocene and underwent progressive infilling driven by climatic fluctuations from the late Holocene to the present (Agarwal et al. 1992; Singh 1996). For the understanding of these Holocene environmental and climate change events at the regional to global scale, pollen analysis has become one of the common proxies (Zhao et al. 2009). However, the specific palaeoenvironmental history of Assam in Brahmaputra plains remains inadequately understood.

The Brahmaputra River, the seventh largest river in the world (Hovius 1998; Tandon & Sinha 2007), has formed a vast and extensive valley in the state of Assam, India. With a sediment yield of 852.4 t/km²/year, the Brahmaputra River holds the highest sediment yield globally (Latrubesse 2008). The Guwahati city of Assam, is a fast growing city situated at the bank of river Brahmaputra.

The present study has been carried out in pursuit of understanding the vegetation and climate change in the recent past in Northeast India. Although the absolute dates for the sample collected could not be procured due to some constraints, nonetheless the vegetation changes obtained from the core samples and associated climate change are discerned and discussed based on relative chronology established for the present study, on the basis of available studies of sediment cores having radiometric chronology from nearby areas. This study contributes to the growing body of palaeoenvironmental research and provides a foundation for future investigations incorporating absolute dating techniques to further refine our understanding of vegetation and climate dynamics in the recent past.

REGIONAL SETTINGS

Study Site and Geological Setting

The study area is located within the Gauhati University Campus (GUC; 26°9'16.16"N, 91°40'4.66"E), situated on the left bank of the Brahmaputra River in the Kamrup Metropolitan District, Assam, India (Figure 1). The local geology of the area represents an extension of the Shilong Plateau, characterized by Precambrian crystalline basement rocks (Özcan et al. 2018). The major rock types in the region are of Precambrian-Cambrian age and primarily comprise quartz-feldspathic gneiss, granite, granodiorite, amphibolite, and calc-silicate rocks (Shukla et al. 1989). The granitic hills within the Gauhati University campus trend in an east-west direction. The lowland areas where sediment accumulates are covered with brown and reddish sandy loam soils rich in both macro and micro nutrients (Kar & Borthakur 2007).

Climate

The study area experiences a subtropical monsoon climate (Neog & Khinchi 2015) with distinct seasonal variations. Winters (November-February) are characterized by dry and cool conditions, while summers (May-August) are hot and humid. The precipitation regime begins in April with sporadic rainfall events, occasionally manifesting as heavy downpours accompanied by cyclonic disturbances. The South-West Monsoon, contributing approximately 80% of the annual precipitation, typically commences in early June and continues through October with periods of intense rainfall. Long-term meteorological data (1951-2000) indicate that August records the highest average maximum temperature of 32.1°C, while January exhibits the lowest average minimum temperature of 10.2°C. The region receives an average annual precipitation of 1500 mm, predominantly concentrated during the monsoon months (Neog & Khinchi 2015).

Vegetation

The GUC is enriched with a rich biodiversity. Out of the 484 acres of total area, nearly 40% of the campus are marshlands, 10% are under hillocks and rest of the plains are undulating highlands (Sustainable Campus 2021). The vegetation of

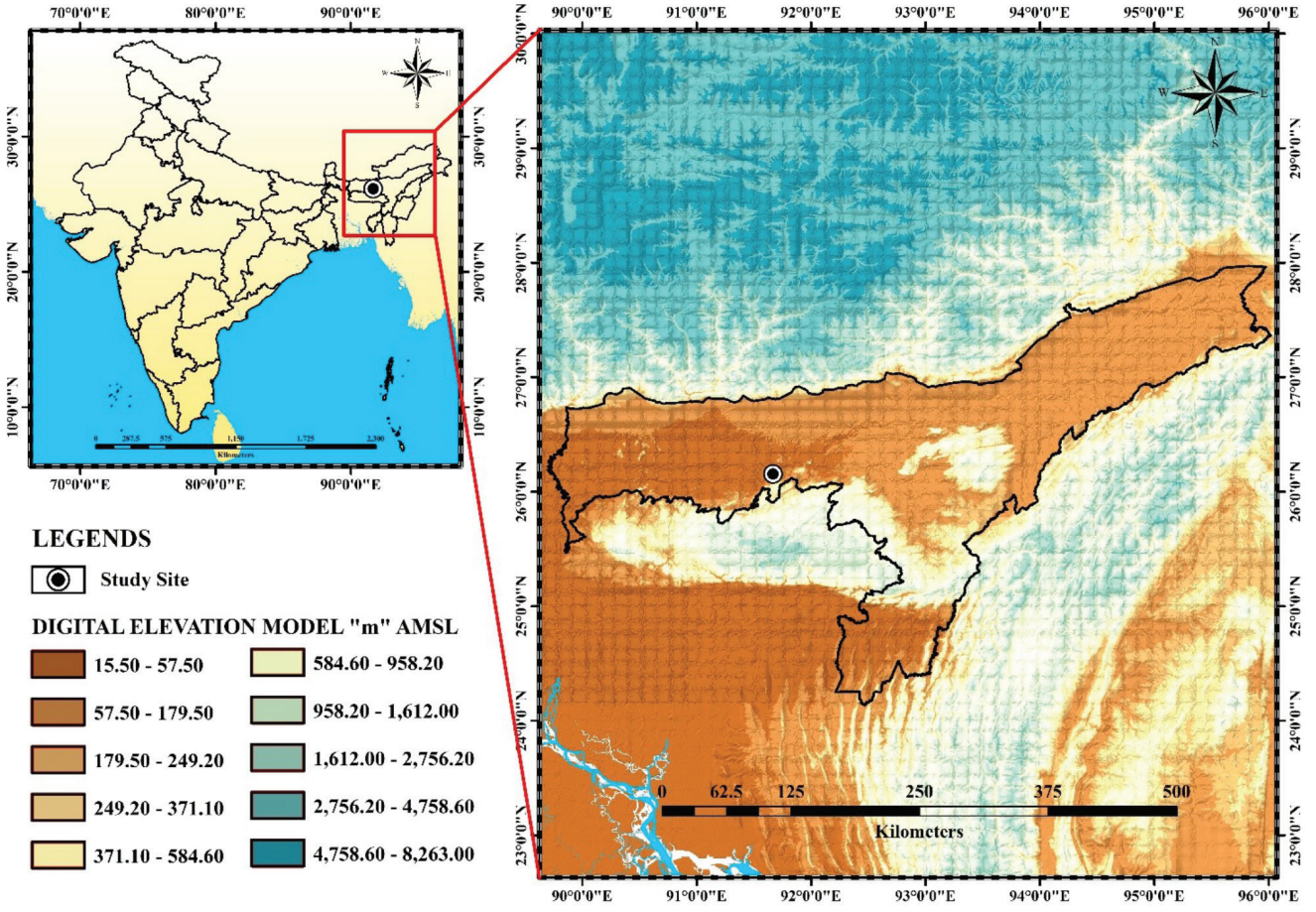


Figure 1. Digital Elevation Model (DEM) illustrating the location of the Gauhati University swamp, Assam, India.

Gauhati University can be broadly classified into five types (Kar & Borthakur 2007) including degraded alluvial inundated grasslands, riparian seasonal carpet vegetation, aquatic vegetation, waste land vegetation and highland vegetation. The Campus comprises about 337 species of Angiospermic taxa wherein monocots are represented by 93 species and dicots by 244 species. *Asteraceae* with 17 species is most dominating family along with *Euphorbiaceae, Verbenaceae, Poaceae, Cyperaceae, Hydrocaritaceae* are other dominating families found in the campus (Kar & Borthakur 2007) (Table 1).

MATERIALS AND METHODS

Sediment trench

For the present palynological investigation at GUC, sediment samples were obtained from a swamp situated within the campus premises. Core extraction was carried out using a 50 cm long PVC pipe. A total of 10 samples were systematically collected from the core for subsequent palynological analysis.

Pollen analysis

For pollen analysis, 10 samples were extracted from the 50 cm deep core, with a consistent sampling interval of 5 cm. After the collection of the samples, about 10 gm of each sample was treated with 10% aqueous KOH to remove the humus. Thereafter, the samples were treated with hydrofluoric acid (HF) in order to remove the silica followed by the standard procedure of acetolysis (Erdtman 1943), using a mixture of acetic anhydrite and conc. sulfuric acid (H_2SO_4)

Table 1. Modern vegetation across the study region.

S. No.	Species name	Family
1	*Albizia lebbeck*	*Fabaceae*
2	*Albizia lucidior*	*Fabaceae*
3	*Alstonia scholaris*	*Apocynaceae*
4	*Alternanthera philoxeroides*	*Amaranthaceae*
5	*Areca catechu*	*Arecaceae*
6	*Artocarpus chaplasha*	*Moraceae*
7	*Artocarpus heterophyllus*	*Moraceae*
8	*Azadirachta indica*	*Meliaceae*
9	*Bombax ceiba*	*Malvaceae*
10	*Caryota urens*	*Arecaceae*
11	*Cassia fistula*	*Fabaceae*
12	*Chukrassia tabularis*	*Meliaceae*
13	*Cocos nucifera*	*Arecaceae*
14	*Colocasia esculenta*	*Araceae*
15	*Delonix regia*	*Fabaceae*
16	*Dillenia indica*	*Dilleniaceae*
17	*Diplazium esculentum*	*Athyriaceae*
18	*Erythrina indica*	*Fabaceae*
19	*Ficus religiosa*	*Moraceae*
20	*Ficus rumphii*	*Moraceae*
21	*Ipomoea aquatica*	*Convolvulaceae*
22	*Lagerstroemia speciosa*	*Lythraceae*
23	*Lannea coromandelica*	*Anacardiaceae*
24	*Ludwigia octovalvis*	*Onagraceae*
25	*Mangifera indica*	*Anacardiaceae*
26	*Melia azedarach*	*Meliaceae*
27	*Mesua ferrea*	*Calophyllaceae*
28	*Michelia champaca*	*Magnoliaceae*
29	*Millettia pinnata*	*Fabaceae*
30	*Mimusops elengi*	*Sapotaceae*
31	*Monoon longifolium*	*Annonaceae*
32	*Neolamarckia cadamba*	*Rubiaceae*
33	*Oroxylum indicum*	*Bignoniaceae*
34	*Parthenium hysterophorus*	*Asteraceae*
35	*Peltophorum pterocarpum*	*Fabaceae*
36	*Phoenix dactylifera*	*Arecaceae*
37	*Pontederia crassipes*	*Pontederiaceae*
38	*Pteris* spp.	*Pteridaceae*
39	*Samanea saman*	*Fabaceae*
40	*Saccharum* spp.	*Poaceae*(grass)
41	*Syzygium cumini*	*Myrtaceae*
42	*Syzygium jambos*	*Myrtaceae*
43	*Setaria* spp.	*Poaceae*(grass)
44	*Terminalia chebula*	*Combretaceae*

in the ratio of 9:1. In the end, the samples were preserved in 50% glycerin and few drops of phenol solution for microscopic examination. Pollen and spore identification was carried out based on prior studies (Basumatary et al. 2014; Trivedi et al. 2024) and by comparing the specimens with those in the pollen slides collection at the herbarium of the Birbal Sahni Institute of Palaeosciences. Temporary pollen slides were prepared, and pollen and spore identification data was documented using an Olympus BX52 microscope equipped with an Olympus DP23 camera at a magnification of ×400.

RESULTS

The palynological record procured from the GUC site core, was divided into three distinct pollen zones on the basis of characteristic palyno-composition and quantification of major arboreal and non-arboreal pollen taxa (Figure 2). The chronological sequence ranges from Pollen Zone GU-I, representing the oldest sediments, to Pollen Zone GU-III, corresponding to the most recent sediments. Some significant pollens and spores are depicted in Figure 3.

Pollen Zone GU-I (50-30 cm): *Syzygium cumini, Ailanthus excelsa*, *Bombax ceiba, Pinus, Abies, Shorea robusta, Poaceae*, *Cerealia*

This pollen zone is characterized by a dominance of arboreal pollen (AP) over non arboreal pollen (NAP). Among AP, prominent highland tree taxa such as *Abies* (1.5-4.2%), *Pinus* (1.9-5.9%), and *Alnus* (0.7%) are well represented in this zone. Moist deciduous taxa such as *Syzygium cumini* (15.9-28.2%) dominates this zone with a high frequency followed by *Ailanthus excelsa* (6.7-22.4%). Other arboreal taxa include *Myrtaceae* (5.9-6.7%), *Shorea robusta* (0.7-2.5%), *Lagerstroemia* (1.18%), *Bombax ceiba* (4.6%), *Palmae* (0.84%), *Anacardaceae* (9.3-14.2%). NAP are dominated by terrestrial herbs including *Poaceae* (10.7-20.7%) followed by *Cerealia* (2.9-8%) and other potential cultivated taxa such as *Chenopodiaceae* (0.8-1.4%) and *Caryophyllaceae* (0.5-2.9%). The heathland taxa are represented by *Asteraceae* (2.1-5%), *Ageratum conyzoides* (2.2-2.9%), *Aster tripolium* (0.6-

4.4%) and *Alternanthera* (0-2.5%). Additional terrestrial herbs include *Eclipta alba* (0-4.2%) and *Convolvulceae* (0-0.8%). Ferns are also prominent in this zone, with a frequency of 11.4-18.6%.

Pollen Zone GU-II (30-15 cm): *Syzygium cumini, Pinus, Ailanthus excelsa, Anacardaceae, Poaceae, Cerealia*

This zone is marked by a predominance of AP over NAP. The frequencies of highland taxa such as *Pinus* (2.4-11.6%), *Abies* (0.4-3.2%) and *Alnus* (0.4-0.6%) are higher compared to previous zone. Among the moist deciduous taxa *Syzygium cumini* (16.2-32.1%) continues to dominate followed by Myrtaceae (7.1-17.3%), and *Ailanthus excelsa* (1.9-6.4%). *Shorea robusta* (0.7%) and *Bombax ceiba* (0.7%), are present in minor quantities, while *Anacardaceae* (7.9-16.2%) shows increased percentage. NAP such as *Poaceae* (7.4-9.7%) declines from previous phase while *Cerealia* (1.2-12.8%) increases along with potential cropland taxa such as *Caryophyllaceae* (4.5-9.2%) show an increase in abundance, along with *Chenopodiaceae* (0.7%) as compared to previous zone. The heathland taxa viz., *Asteraceae* (11.6%), *Ageratum conyzoides* (0-2.5%), *Aster tripolium* (5.8%), *Alternanthera* (4.9%) are present in scattered quantities. *Ludwigia* (4.9%) a marshy taxon appears for the first time in this zone. More terrestrial including *Eclipta alba* (1.4%) reduced and *Convolvulceae* disappeared in this zone. Ferns (20.7-29.7%) are notably more abundant compared to the previous zone.

Pollen Zone GU-III (15-0 cm): *Syzygium cumini, Alianthus excelsa, Pinus, Shorea robusta, Bombax ceiba, Poaceae, Cerealia*

During this phase there is a significant increase in AP than NAP. This zone is characterized by the dominance of AP than NAP. Among AP, moist deciduous taxa such as *Syzygium cumini* (11.1-26.2%), *Myrtaceae* (11.1-13.5%), *Alianthus excelsa* (12.2-21.7%) are prominent, showing higher frequencies compared to earlier zones. Others species including, *Shorea robusta* (0.5-1.65%), *Lagerstroemia* (16%), *Bombax ceiba* (0.5-1.1%) also showed increased abundance. *Palmae* (1-1.6%) reappeared in this zone. Among the highland taxa's *Pinus* (4-7%), *Abies* (0.5%) and *Alnus* (0.5%) show lower frequencies than previous zone. NAP including *Poaceae* (3.8-4.8%) and Cerealia (5.9-6.1%) are declining significantly in this zone when compared to the earlier phases. The other non-arboreals viz., *Caryophyllaceae* (1.6-2.7%) also exhibit reduced values. *Anacardaceae* (10.8-25%) is the only shrub represented. More terrestrial herbs include *Eclipta alba* (0.5%), and *Convolvulceae* (1-1.1%). The heathland taxa such as *Aster tripolium* (1.11%), *Asteraceae* (4.3-6.1%), and *Alternanthera* (1.6%) are in moderately low values while *Ageratum conyzoides* disappeared in this zone. The marshy taxa *Ludwigia* (0.5-1.1%) reduces significantly. Fern spores (27.4-46.1%) are more abundant in this zone, with higher frequencies compared to the previous two zones.

DISCUSSION

Climate serves as a major determinant influencing plant distribution patterns thereby playing a critical role in the reconstruction of past vegetation dynamics and associated climatic conditions. In the present study, pollen assemblages retrieved from the swamp within Gauhati University in Assam has been employed to infer past vegetation structure and climate variability. As mentioned earlier, the absolute chronological control is not available for the studied core, however, the relative ages are estimated on the basis of other available palynological studies of the sediment cores. The core lengths and chronologies of these cores are plotted and vegetation and climate inferences are envisaged on the basis of estimated chronology (Figure 4). The retrieved palyno-composition is categorized into three distinct zones, and is instrumental in understanding the vegetation dynamics that had occurred in the recent past in the studies region.

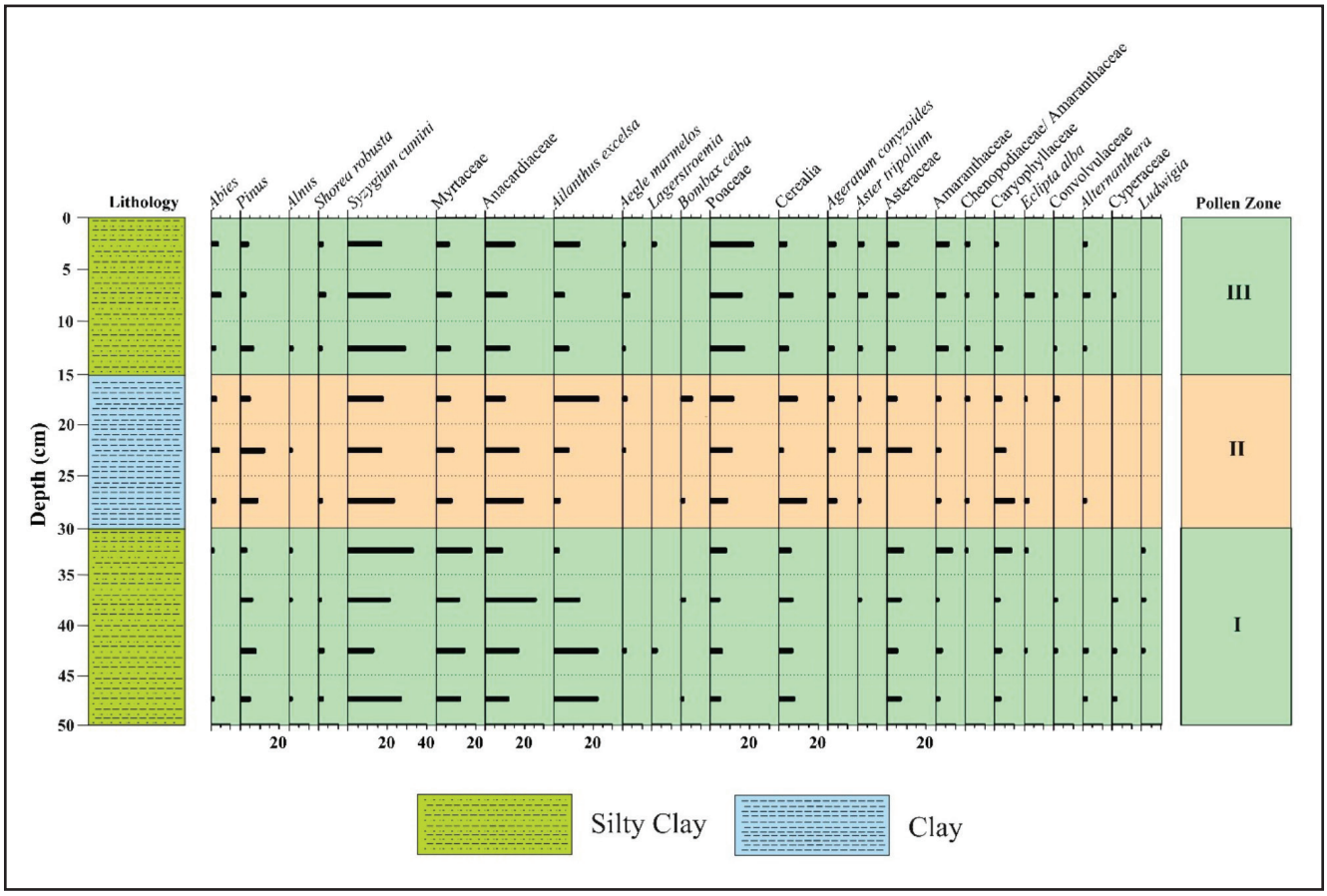


Figure 2. Pollen frequency chart recovered from the Gauhati University swamp, Assam, India.

The salient features of each zones are discussed in the following.

The palynological analysis of the core sediments corresponding to Pollen Zone GU-I (50-30 cm) indicates a notable expansion of forest grooves under a warm and humid climatic regime, accompanied by increased precipitation. The presence of *Shorea robusta* signifies that the region was receiving good amount of precipitation and timely onset of monsoon. Additionally, the presence of *Cerealia* alongside potential cultivated taxa *Chenopodiaceae* and *Caryophyllaceae* points towards agricultural activities in the vicinity of the swamp during this phase. These findings collectively suggest an intensification of the ISM during this period, consistent with the Holocene Climate Optimum (HCO) observed globally between 8000 and 4000 cal yr BP (Benarde 1996). Comparable records from Renuka Lake in the Northwestern Himalaya also document the expansion of moist deciduous forests, including *Shorea robusta*, indicative of strengthened monsoon conditions during the mid-Holocene, coinciding with the HCO (Nag et al. 2025). Similarly, multiple sites across Northeast India including lakes in the Mikir Hills (Bera 2003; Trivedi 2022), Dabaka Swamp (Dixit & Bera 2012), Chayagaon Swamp (Dixit & Bera 2013), Srinagar Lake (Bhattacharyya et al. 2011), and Ziro Lake (Ghosh et al. 2014) also reflect enhanced precipitation associated with the HCO (Figure 4). Further supporting evidence comes from the Central Ganga Plain, where records from Lahuradewa Lake (Chauhan et al. 2009), Jalesar Lake (Trivedi et al. 2013), Kikar Tal (Saxena & Trivedi 2017), and Lashoda Tal (Trivedi et al. 2019), indicate warm and humid conditions and an active southwest monsoon during the same interval.

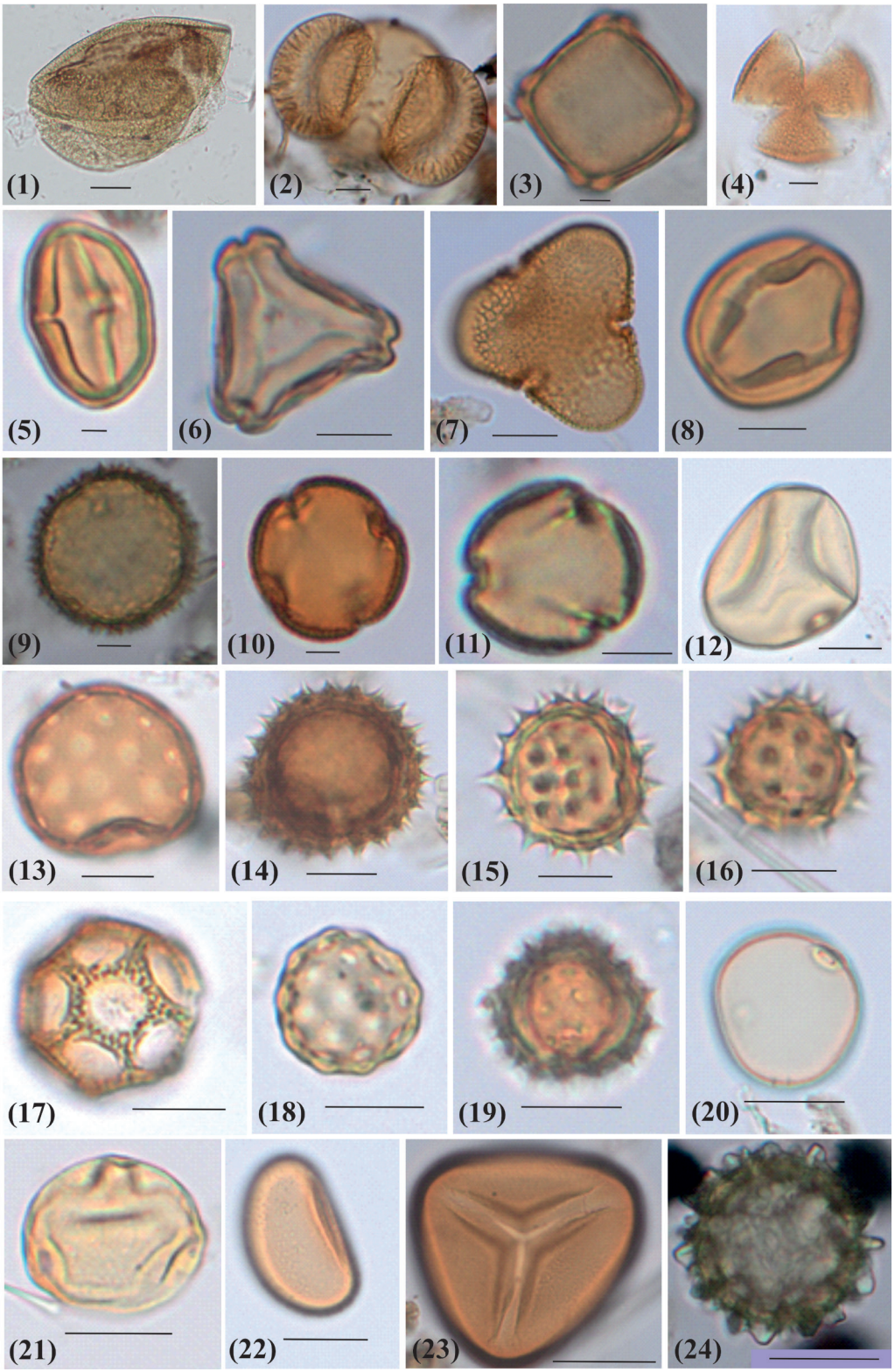


Figure 3. Some pollen and spores recovered from the Gauhati University swamp. 1. *Abies* 2. *Pinus* 3. *Alnus* 4. *Shorea robusta* 5. *Lagerstroemia* 6. *Eucalyptus* 7. *Anacardiaceae* 8. *Bombax ceiba* 9. *Convolvulaceae* 10. *Aegle marmelos* 11. *Ailanthus excelsa* 12. Cerealia 13. *Caryophyllaceae* 14. *Eclipta alba* 15. *Asteraceae* 16. *Aster tripolium* 17. *Alternanthera* 18. *Chenopodiaceae/Amaranthaceae* 19. *Xanthium* 20. *Poaceae* 21. *Ludwigia* 22. Fern Monolete 23. Fern Trilete 24. Algal spore.

In Pollen Zone GU-II (30-15 cm), a decline in forest groves and a corresponding expansion of open landscapes with scattered vegetation is evident. The reduced abundance of *Shorea robusta*, compared to the preceding phase, suggests that the region experienced less warm and marked decline in precipitation. A concurrent decrease in *Bombax ceiba* among the arboreal taxa further supports the inference of diminishing rainfall. Moreover, the notable increase in *Cerealia* and *Caryophyllaceae* indicates intensification of agricultural activities in the vicinity of the swamp. The first appearance of marshy taxa such as *Ludwigia* point towards the development and stabilization of marshy fringes along the periphery of the swamp, further suggesting a contraction of the lake extent during this phase.

Similar evidence has been documented from the Deosila Swamp in Northeast India, indicating that cool and dry climatic conditions prevailed between 6340 and 2970 cal yr BP (Dixit & Bera 2011). Records from Renuka Lake also reflect a weakening of the ISM between 4460 and 3480 cal yr BP, suggesting the dominance of cool and dry conditions during this interval (Nag et al. 2025). Additionally, speleothem $\delta^{18}O$ records from Mawmluh Cave identified a drought phase between 4200 and 3900 cal yr BP (Kathayat et al. 2018). Pollen-based vegetation and moisture reconstructions from Tso Moriri Lake in Ladakh reveal a long-term Holocene trend toward increasing aridity from ~5200 to 3000 cal yr BP, with particularly heightened dryness between ~4500 and 3600 cal yr BP (Leipe et al. 2014). Collectively, these findings support the occurrence of the 4200 cal yr BP event an abrupt, globally

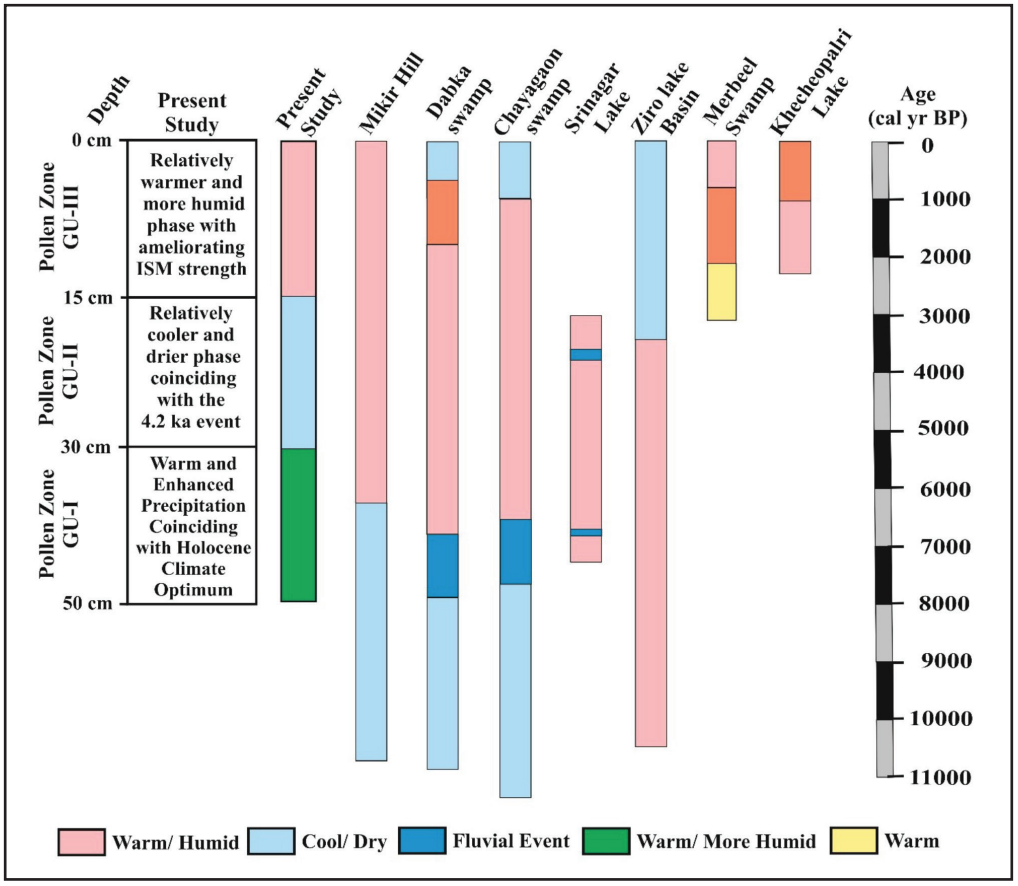


Figure 4. Comparative chart illustrating the relative chronology of selected lake systems across Northeast India and inferences drawn for present study.

recognized cold and dry phase marking the onset of the Meghalayan age (Booth et al. 2005).

In Pollen Zone GU-III (15-0 cm), the pollen assemblage reflects an amelioration in climatic conditions across the region. An increase in tropical moist deciduous taxa, particularly *Shorea robusta*, suggests a shift toward warmer and more humid conditions, accompanied by a timely monsoon onset than previous phase. Furthermore, a decline in the relative abundance of *Cerealia* and other cultivated taxa indicates a reduction in agricultural activity around the swamp. Comparable records from lakes in Northeast India, including the Merbeel Swamp (Bera & Dixit 2011), Khecheopalri Lake (Sharma & Chauhan 1999), and Mikir Hills (Bera 2003), indicate warm and humid conditions associated with enhanced precipitation in their recent sediment profiles. Similarly, records from Renuka Lake suggest the persistence of moderate Indian Summer Monsoon (ISM) conditions during this interval (Nag et al. 2025). Additional evidence from the Kerha Swamp in Madhya Pradesh points to a robust monsoon phase that facilitated the expansion of *Shorea robusta* (Sal) forests, indicative of a warmer, more humid climate and the timely onset of the Southwest Monsoon from ~400 cal yr BP to the present (Chauhan 2005).

CONCLUSION

1. A distinct signal of the Holocene Climate Optimum (HCO) is recorded at Pollen Zone GU-I (50–30 cm depth), marked by the expansion of *Shorea robusta* and other tropical moist deciduous taxa under a strengthened Indian Summer Monsoon (ISM).
2. Palynological evidence from Pollen Zone GU-II (30–15 cm depth) indicates a transition from dense forest cover to an open vegetation landscape, reflecting a weakening of monsoonal strength.
3. The decline in arboreal pollen taxa and emergence of marshy indicators during Pollen Zone GU-II coincides with the global 4200 cal yr BP aridification (Meghalayan age) event, highlighting both regional climatic deterioration and anthropogenic activity impacting the swamp ecosystem.
4. Recent sediment profiles (Pollen Zone GU-III, 15–0 cm depth) exhibit a partial recovery of moist deciduous elements, including *Shorea robusta*, suggesting an amelioration of climatic conditions in the late Holocene.

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REFERENCES

Agarwal A.K., Rizvi M.H., Singh I.B., Kumar A. & Chandra S. 1992. Carbonate deposits in Ganga Plain. In: Singh, I.B. (Ed.), Gangetic Plain: Terra Incognita. Geology Department, Lucknow University, pp. 35–43.

Basumatary S. K., Bera S. K., Sangma S. N. & Marak G. 2014. Modern pollen deposition in relation to vegetation and climate of Balpakram valley, Meghalaya, northeast India: Implications for Indo-Burma palaeoecological contexts. Quaternary International, 325, 30-40.

Benarde M.A. 1996. Global Warming. John Wiley and Sons, New York, U.S.A.

Bera S.K. & Basumatary S. K. 2013. Vegetation history and monsoonal fluctuations during the last 12,500 years BP inferred from pollen record at Lower Subansiri Basin, Assam, Northeast India. India. Journal of Palaeosciences 62(1-2): 1–10.

Bera S.K. 2003. Early Holocene pollen data from Mikir Hills, Assam, India. Journal of Palaeosciences 52(1-3): 121-126.

Bhattacharyya A., Mehrotra N., & Shah S.K. 2011. Holocene vegetation and climate of south Tripura based on palynological analysis. Journal of the Geological Society of India 77: 521-526.

Booth R.K., Jackson S.T., Forman S.L., Kutzbach J.E., Bettis III E. A., Kreigs J. & Wright D.K. 2005. A severe centennial-scale drought in midcontinental North America 4200 years ago and apparent global linkages. The Holocene 15(3): 321-328.

Breitenbach S. F., Adkins J. F., Meyer H., Marwan N., Kumar K.K., & Haug G.H. 2010. Strong influence of water vapor source dynamics on stable isotopes in precipitation observed in Southern Meghalaya, NE India. Earth and Planetary Science Letters 292(1-2): 212-220.

Chauhan M. S. 2005. Pollen record of vegetation and climatic changes in northeastern Madhya Pradesh during last 1,600 years. Tropical Ecology 46(2): 265.

Chauhan M. S., Pokharia A.K. & Singh I. B. 2009. Pollen record of Holocene vegetation, climate change and human habitation from Lahuradewa lake, Sant Kabir Nagar District, Uttar Pradesh, India. Man and Environment 34(1): 88-100.

Dixit S. & Bera S.K. 2011. Mid-Holocene vegetation and climatic variability in tropical deciduous sal (Shorea robusta) forest of lower Brahmaputra valley, Assam. Journal of the Geological Society of India 77: 419-432.

Dixit S. & Bera S.K. 2012. Holocene climatic fluctuations from Lower Brahmaputra flood plain of Assam, northeast India. Journal of Earth System Science 121: 135-147.

Dixit S. & Bera S.K. 2013. Pollen-inferred vegetation vis-à-vis climate dynamics since Late Quaternary from Western Assam, Northeast India: signal of global climatic events. Quaternary International 286: 56-68.

Ghosh R., Paruya D. K., Khan M. A., Chakraborty S., Sarkar A. & Bera S. 2014. Late Quaternary climate variability and vegetation response in Ziro Lake Basin, Eastern Himalaya: A multiproxy approach. Quaternary International 325: 13-29.

Hovius N. 1998. Controls on Sediment Supply by Large Rivers, Relative Role of Eustacy, Climate, and Tectonism in Continental Rocks, Gary Kocurek. 59. SEPM Society for Sedimentary Geology. https://doi.org/10.2110/pec.98.59.0002.

Kar A. & Borthakur S.K. 2007. Angiospermic flora of Gauhati University Campus, Assam, India. Genus 189(55): 244.

Kathayat G., Cheng H., Sinha A., Berkelhamme, M., Zhang H., Duan P., Li H., Li X., Ning Y. & Edwards R.L. 2018. Evaluating the timing and structure of the 4.2 ka event in the Indian summer monsoon domain from an annually resolved speleothem record from Northeast India. Climate of the Past 14(12): 1869-1879.

Latrubesse E. M. 2008. Patterns of anabranching channels: The ultimate end-member adjustment of mega rivers. Geomorphology 101(1-2): 130-145.

Leipe C., Demske D., Tarasov P. E. & Members H.P. 2014. A Holocene pollen record from the northwestern Himalayan lake Tso Moriri: implications for palaeoclimatic and archaeological research. Quaternary International 348: 93-112.

Nag A., Trivedi, A., Farooqui, A. & Morthekai, P. 2025. Palaeoclimatic Signatures Based on Pollen Fingerprints: Reconstructing Mid–Late Holocene Climate Dynamics in Northwestern Himalaya, India. Quaternary 8(1): 6. https://doi.org/10.3390/quat8010006.

Neog D. & Khinchi S.S. 2015. Climate of Assam: Some Significant Analysis. Global Climate Change and Biodiversity, V.L. Media Solutions, New Delhi, 71-84. ISBN: 978-93- 85068-73-7.

Özcan E., Pignatti, J., Pereira, C., Yücel, A. O., Drobne, K., Barattolo, F. & Saraswati, P. K. 2018. Paleocene orthophragminids from the Lakadong Limestone, Mawmluh Quarry section, Meghalaya (Shillong, NE India): implications for the regional geology and paleobiogeography. Journal of Micropalaeontology 37(1): 357-381.

Saxena A. & Trivedi A. 2017. Pollen based vegetation and climate change records deduced from the lacustrine sediments of Kikar Tal (Lake), Central Ganga Plain, India. Journal of Palaeosciences, 66(1-2)): 37-46.

Sharma C. & Chauhan M.S. 1999. Palaeoclimatic inferences from quaternary palynostratigraphy of the Himalayas. The Himalayan Environment. New Age International, New Delhi, 193-207.

Shukla R.C., Gopalkrishnan R. & Chakrabarty S. 1989. Study of Granitic rocks around Kamakhya Hill and adjoining area, Kamrup District Assam, GSI Record. -122 Pt IV P: 72-73.

Singh I.B. 1996. Geological Evolution of Ganga Plain–an Overview. Journal Palaeontological Society of India 41: 99–137.

Sustainable Campus 2021. Gauhati Univrsity portal. Available at: https://web.gauhati.ac.in/landscapes/sustian [Accessed on: 28 April 2025].

Tandon S.K. & Sinha R. 2022. Geology of large river systems. Large Rivers: Geomorphology and Management, Second Edition, 7-41.

Trivedi A. 2022. Holocene vegetation, climate, and culture in Northeast India: a pollen data based review. In: Holocene Climate Change and Environment Elsevier Publishing Group (611-628).

Trivedi A., Agrawal S., Sharma A., Ali S.N., Manoj M.C., Nag A., Misra S. & Kawsar M. 2024. Climatic Oscillations and Dynastic Trends: A Multiproxy analysis of the past two millennia in the Indian Subcontinent. Catena 246: 108424.

Trivedi A., Chauhan, M. S., Sharma, A., Nautiyal, C. M. & Tiwari, D.P. 2013. Record of vegetation and climate during late Pleistocene–Holocene in Central Ganga Plain, based on multiproxy data from Jalesar Lake, Uttar Pradesh, India. Quaternary International 306: 97-106.

Trivedi A., Saxena A., Chauhan M.S., Sharma A., Farooqui A., Nautiyal C.M. Yao Y.F., Wang, Y.F., Li C.S. & Tiwari D.P. 2019. Vegetation, climate and culture in Central Ganga plain, India: A multi-proxy record for Last Glacial Maximum. Quaternary International 507: 134-147.

Zhao Y., Yu Z., Chen F., Zhang J. & Yang B. 2009. Vegetation response to Holocene climate change in monsoon-influenced region of China. Earth Science Reviews 97(1): 242–256.